



ভাৰতীয় প্ৰযুক্তিবিদ্যা প্ৰতিষ্ঠান গুৱাহাটী
भारतीय प्रौद्योगिकी संस्थान गुवाहाटी
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

RT-508

GEOSPATIAL TECHNOLOGY IN AGRO AND RURAL SECTOR



MINI PROJECT:

**Gram Panchayat level utility assessment
(Spatial Assessment)**

PRESENTED BY –

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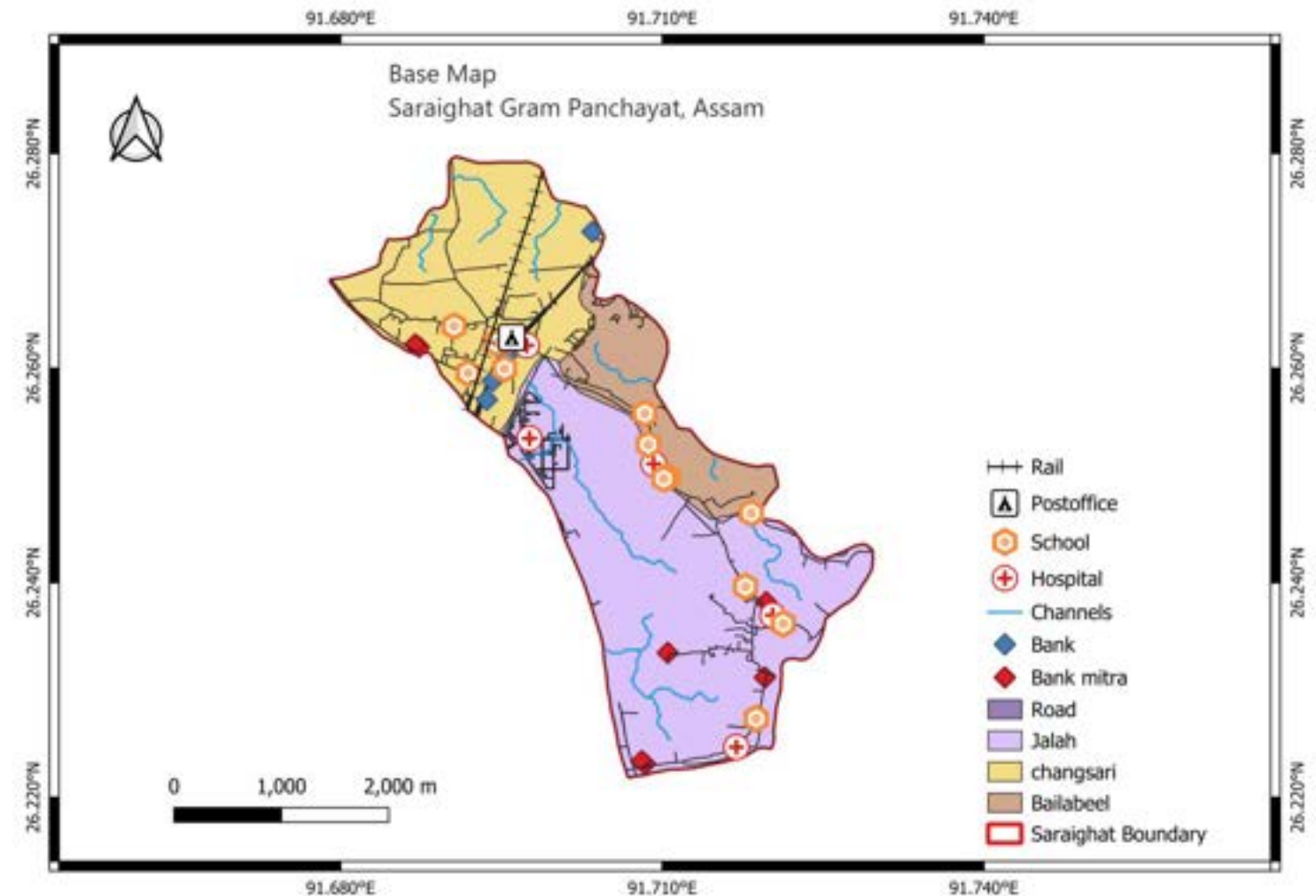
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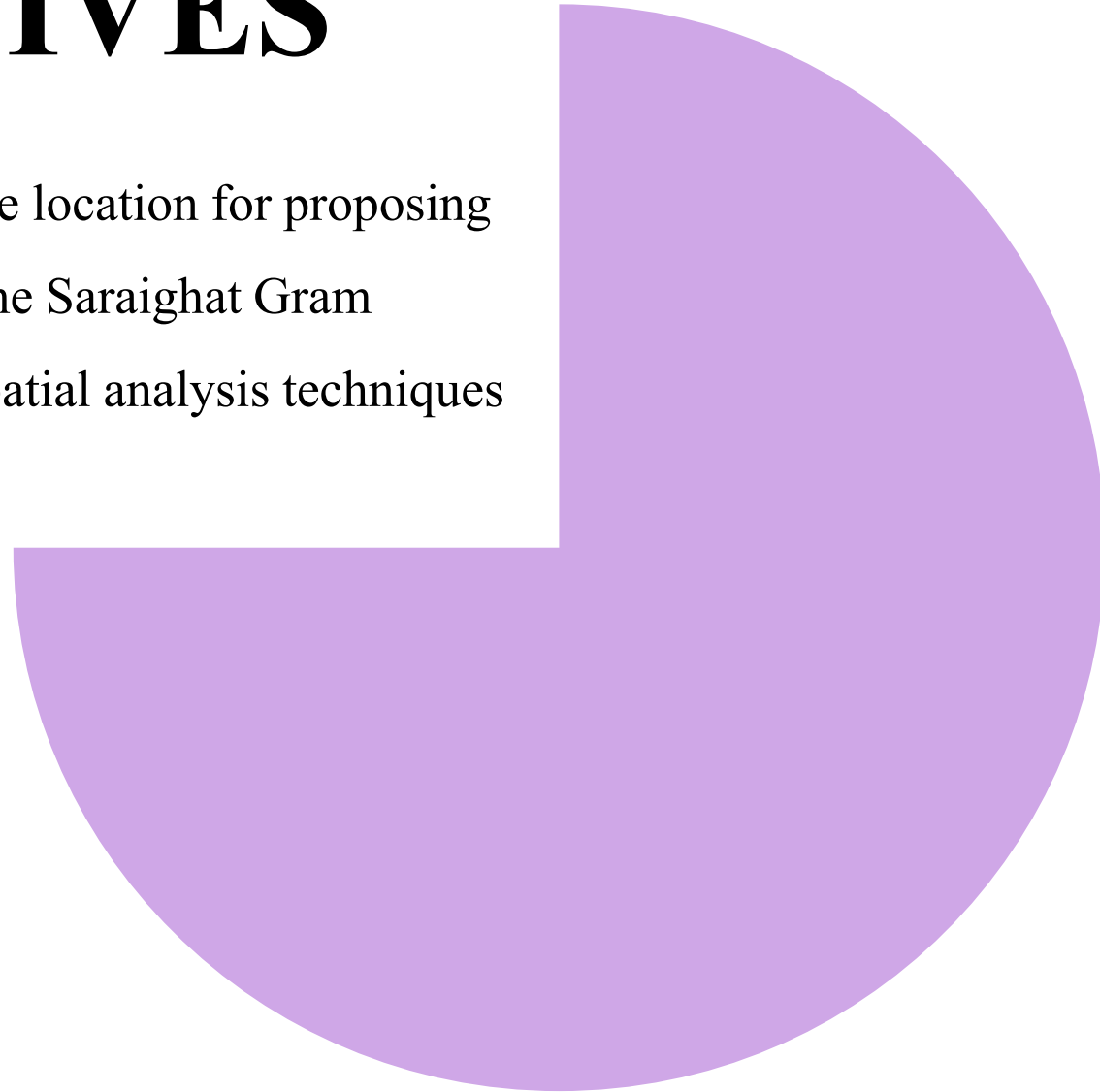
INTRODUCTION

- Saraighat GP is located under Bezera block in Kamrup district of Assam State, India.
- The Saraighat GP consists of three villages : Changsari, Balaibeel, and Jalah.
- The total area is spread over 12.13 sq. km.
- Geographical extent of latitude is $26^{\circ}13' \text{ N}$ – $26^{\circ}17' \text{ N}$ & longitude is $91^{\circ}40' \text{ E}$ – $91^{\circ}44' \text{ E}$.



OBJECTIVES

- To identify the most suitable location for proposing a new hospital site within the Saraighat Gram Panchayat, Assam, using spatial analysis techniques in QGIS



METHODOLOGY

DATA ,SOURCE AND TOOLS

LOCATION:

- Saraighat Gram Panchayat, Assam
- Three village: Changsari, Jalah, Balaibeel
- Width= approx. 2000m, Length = approx. 6000m

UTILITIES:

- Schools 12
- Hospitals 5
- Railway 1
- Banks 9
- Postoffice 1

SOURCE:

- Space-based Information Support for Decentralized Planning (SISDP) Bhuvan Panchayat portal and Bharatmaps NIC portal,
- LULC map temporal satellite data set (Sentinel-2 imagery on 22-10-2024 and 24-02-2025)

TOOLS:

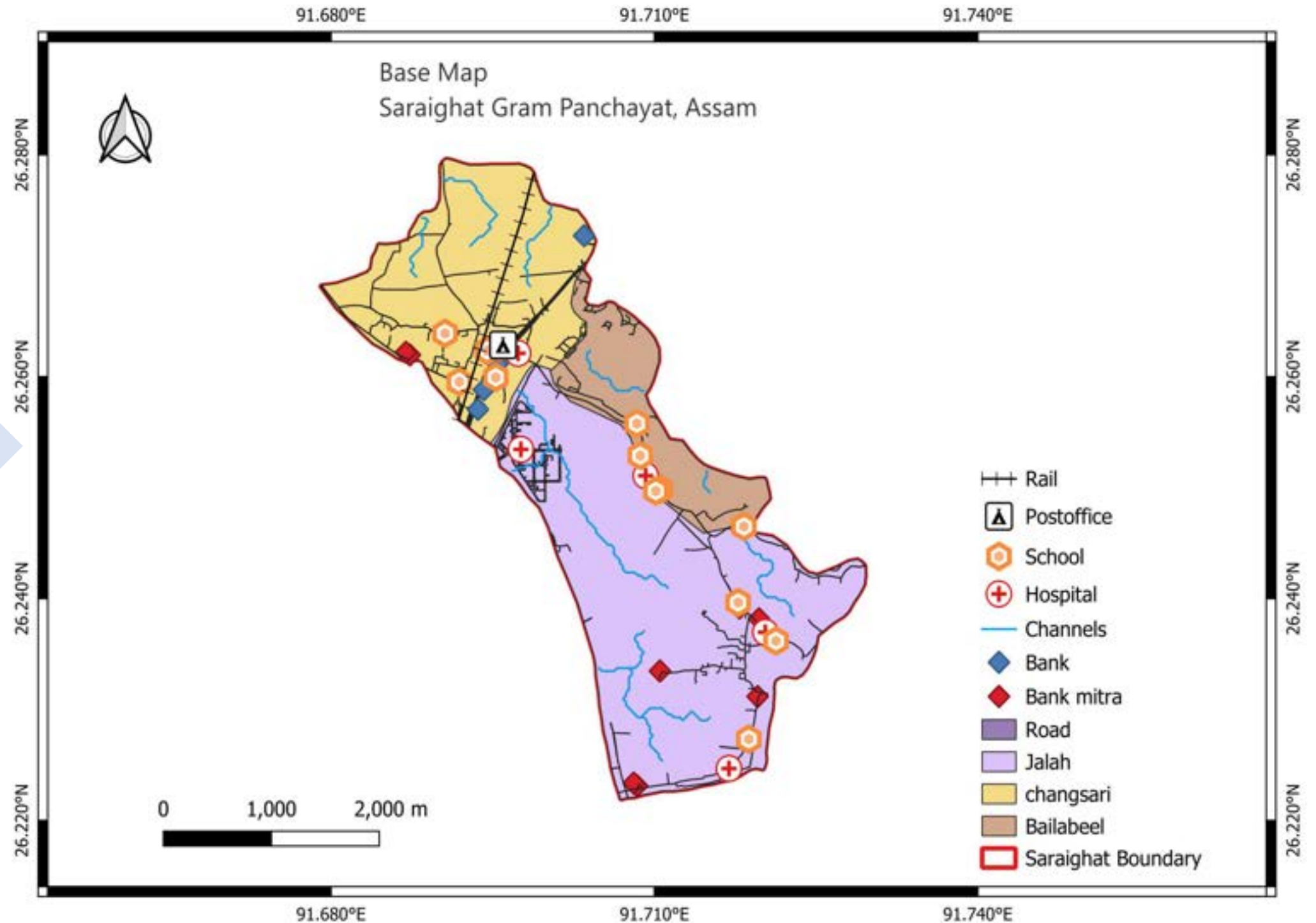
- QGIS software 3.40
- Weighted Multi-Criteria Analysis(WMCA)
- Scale= 1:50000

QGIS Tools

QGIS Tool/Step	Primary Purpose in Suitability Analysis
Buffer	Defines zones of inclusion or exclusion around.
Rasterize (Vector to Raster)	Converts vector data (points, lines, polygons) into a grid/raster format, which is required for cell-by-cell calculations like Weighted Multi-Criteria Analysis.
Fill No Data	Addresses missing or 'No Data' cells in a raster layer by filling them with a specific value (e.g., 0) to prevent calculation errors during subsequent overlays.
LULC & Warp/Reproject	Ensures spatial consistency by standardizing the Coordinate Reference System (CRS), resolution, and extent of all input layers (like Land Use/Land Cover) for accurate overlay.
Weighted Multi-Criteria Analysis (Overlay)	The core analysis: combines prepared, reclassified raster layers, multiplying each by its relative importance (weight), and summing them to create the final suitability map.
Mask Layer (Clip Raster by Mask Layer)	Trims the final suitability raster to the precise boundary of the study area removing extraneous data and making results project-relevant.

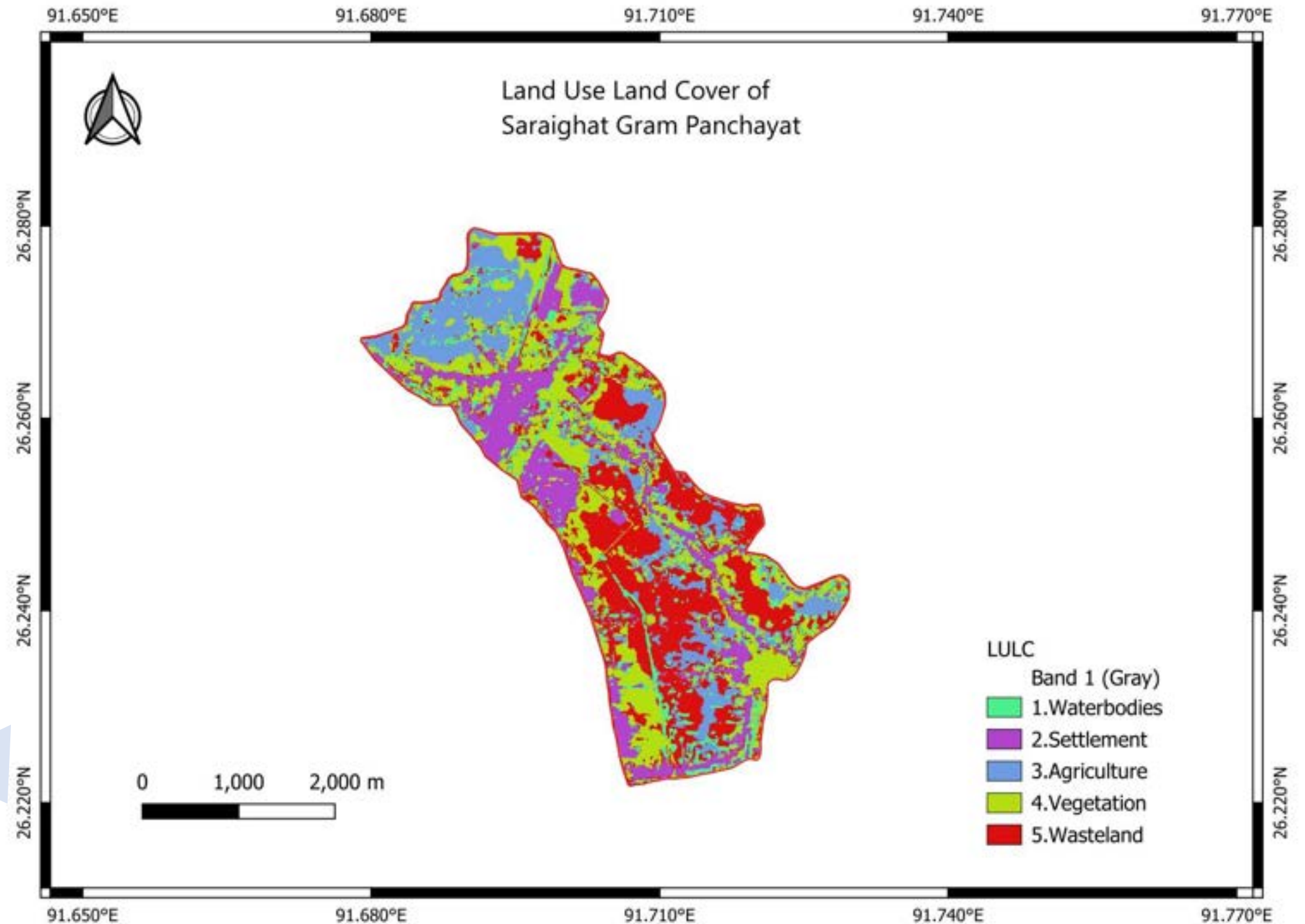
BASE MAP GENERATION

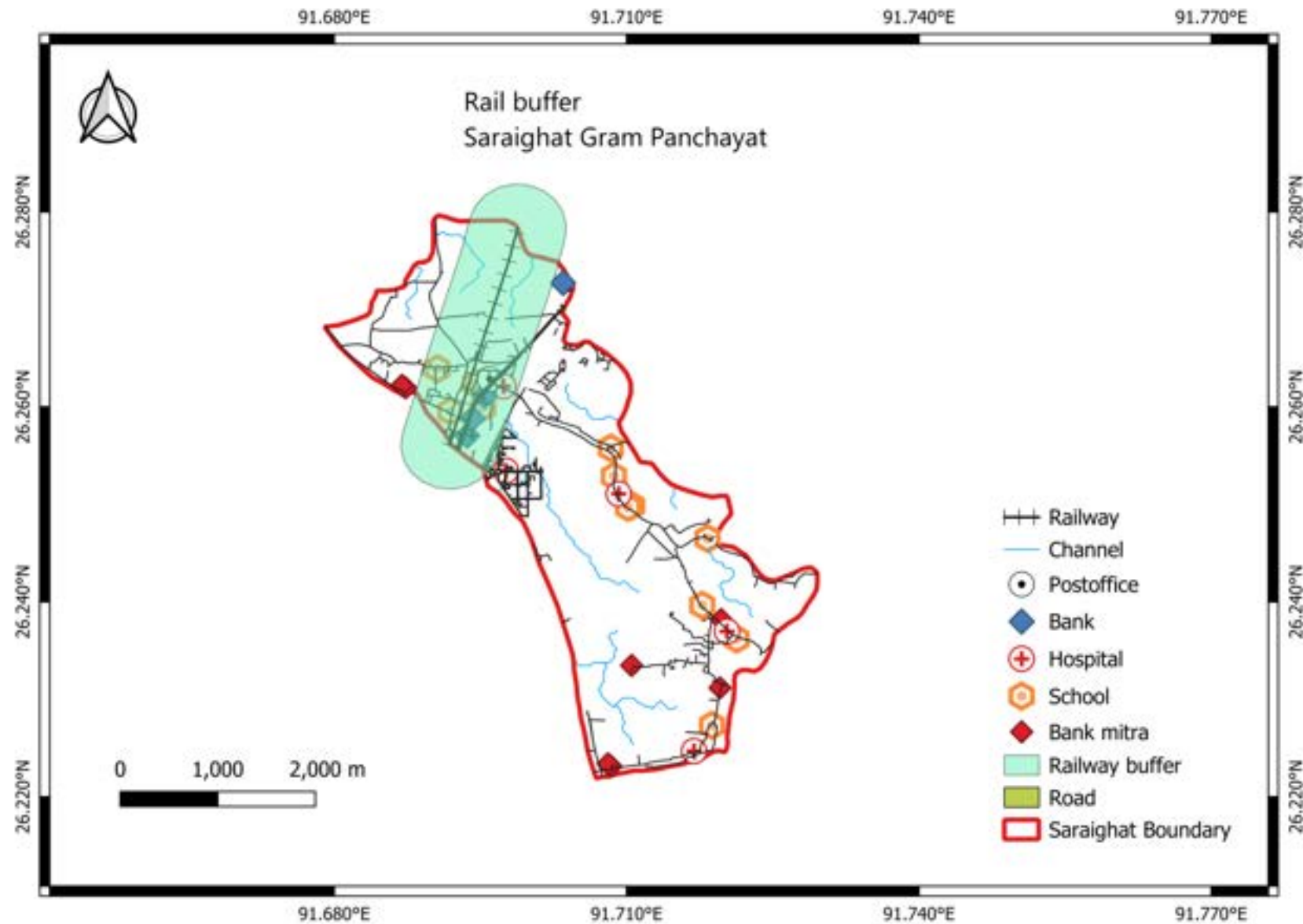
- Insert the utilities shapefiles in QGIS
- Generate a base map for the gram panchayat
- Scale 1:50000



LULC Analysis

- Use the LULC band and analyze the Land Use Land Cover classifications



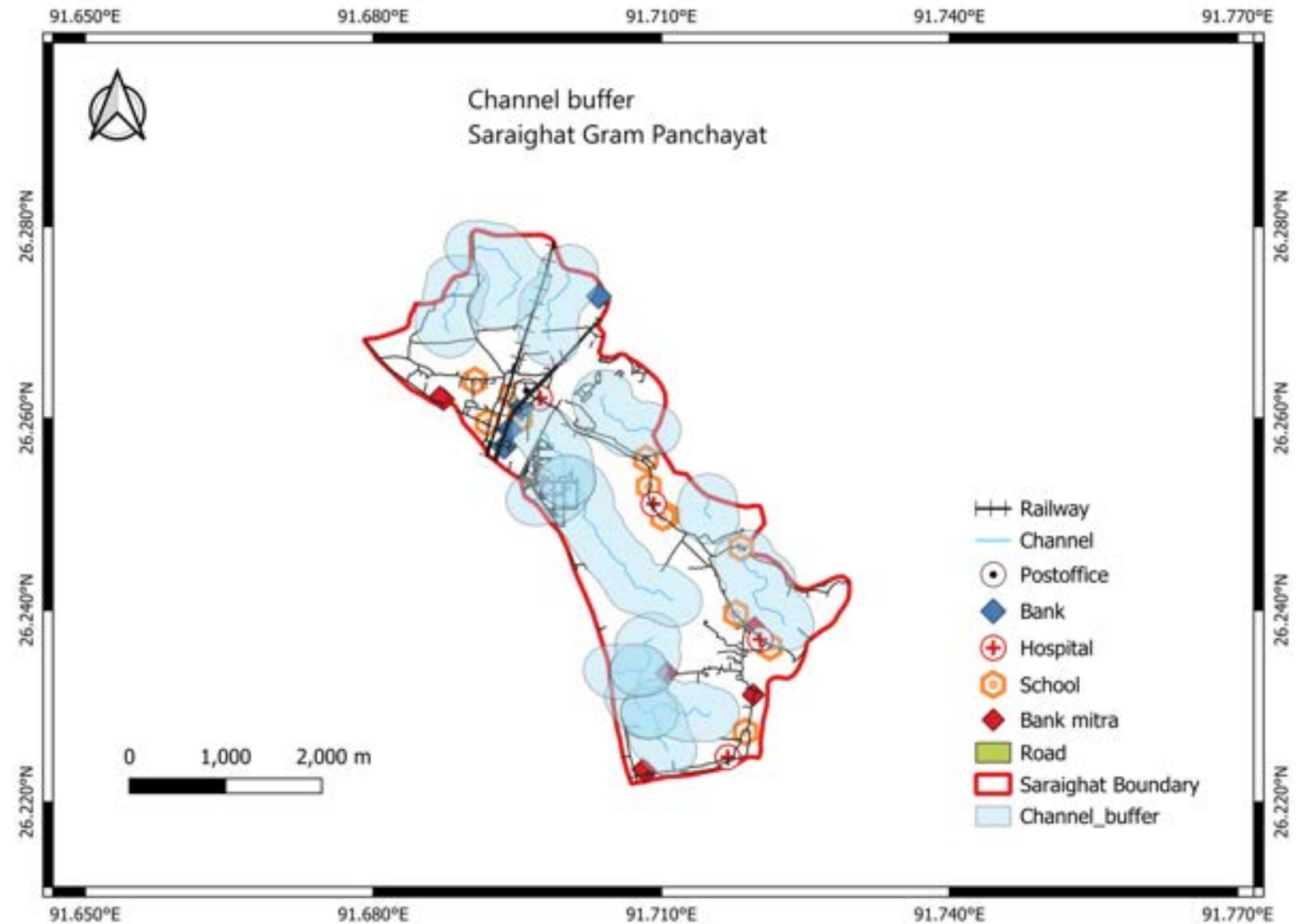


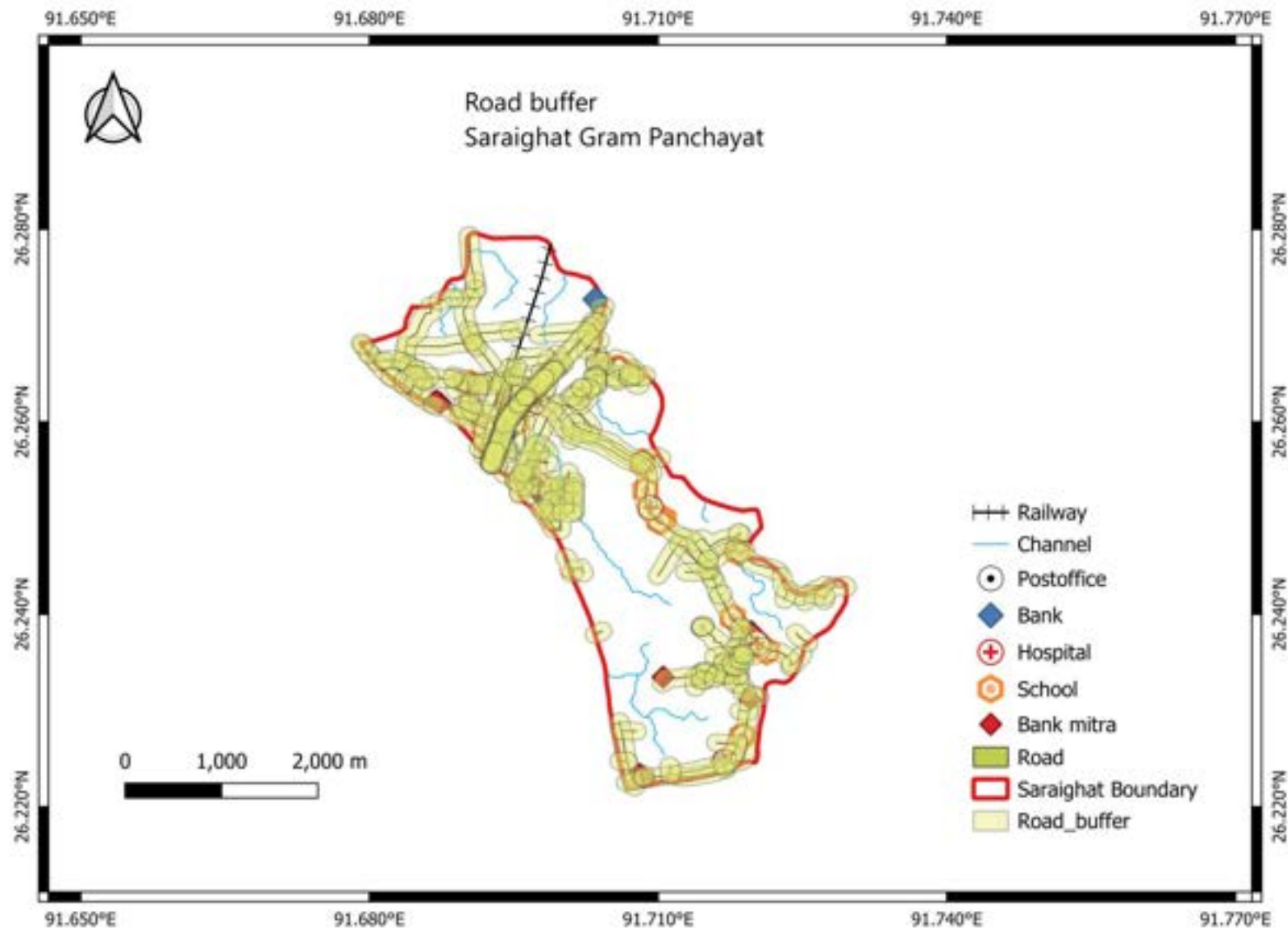
RAILWAY BUFFER

- 500 meter buffer
- Avoidance Constraint: To minimize noise, vibration, and safety hazards, making areas within the buffer unsuitable

CHANNEL BUFFER

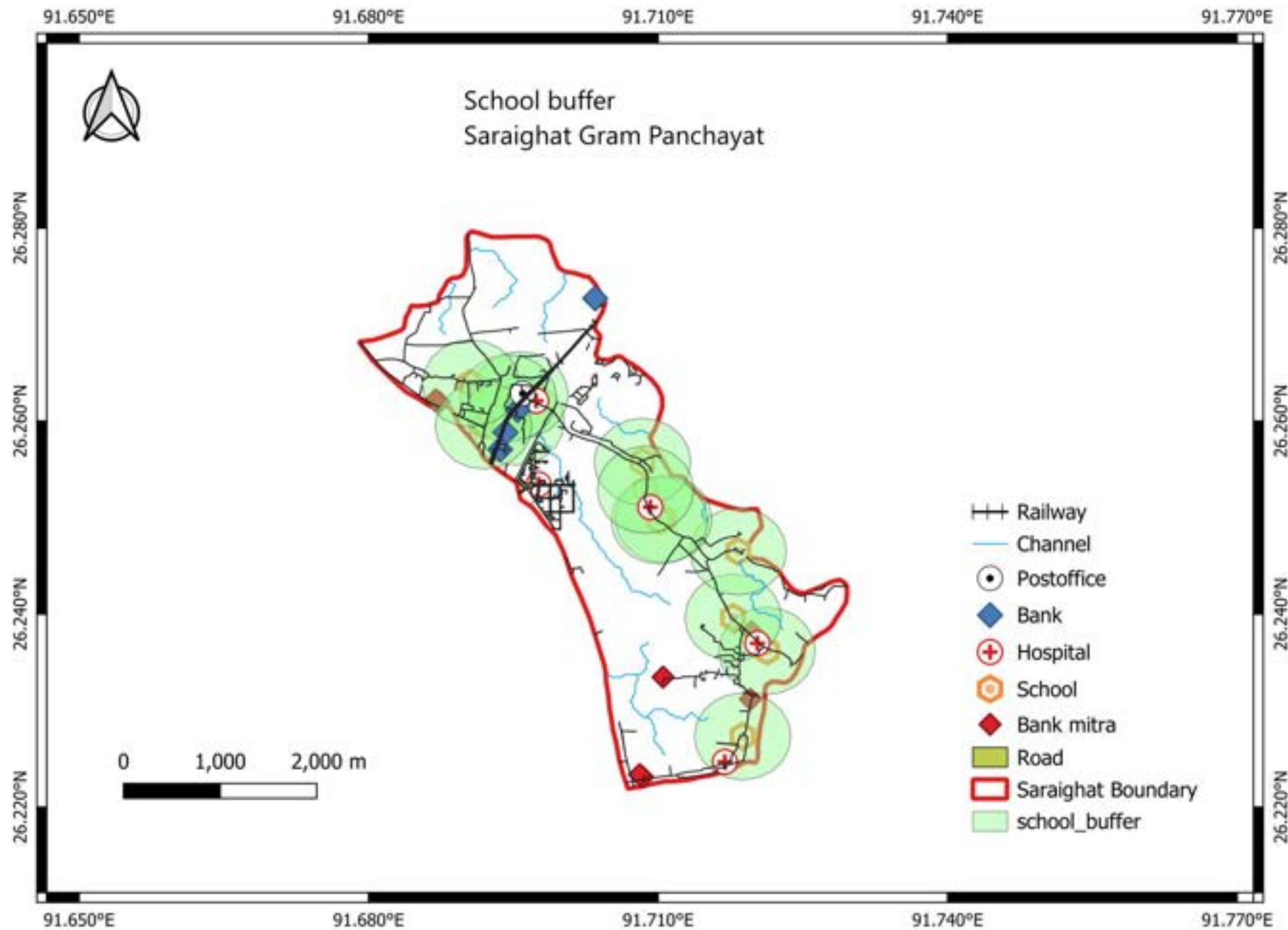
- 300 meter buffer
- Avoidance Constraint:
To prevent construction
in flood-prone areas and
preserve riverbank
zones.





ROAD BUFFER

- 100 meter buffer
- Proximity Requirement: To ensure high accessibility for patients, ambulances, and staff.

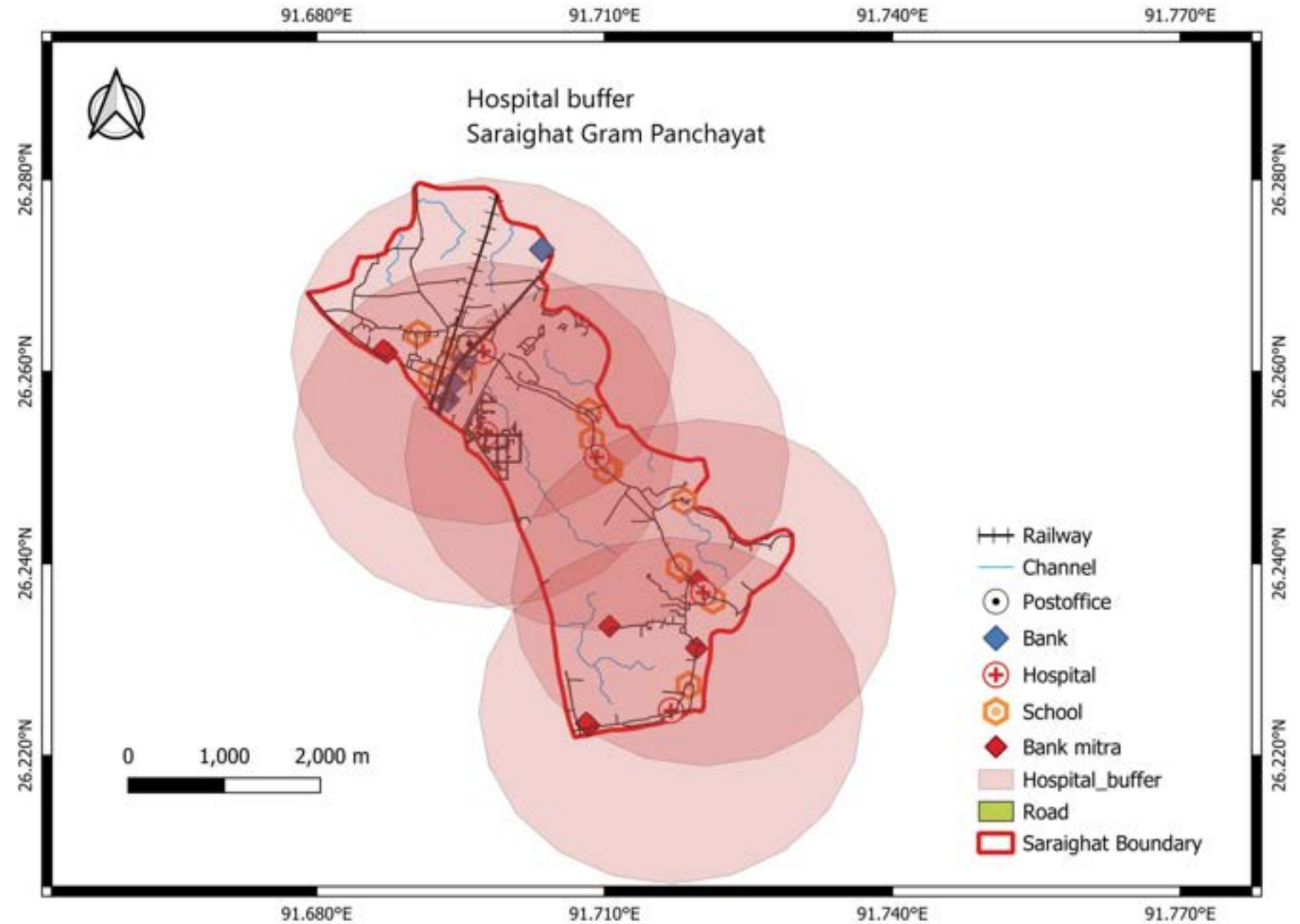


SCHOOL BUFFER

- 500 meter buffer
- Avoidance Constraint: To reduce noise and traffic congestion in educational zones.

EXISTING HOSPITAL BUFFER

- 2000 meter buffer
- Avoidance Constraint: To ensure the new hospital serves an unserved or underserved area (minimizing overlap with existing service areas).



RASTERIZE (Vector to Raster)

- Prepare data for the Weighted Multi-Criteria Overlay.
- All vector buffer layers must be converted to raster format
- This conversion creates a grid of cells (pixels)
- Each cell is assigned a simple pixel value (1 for inside the buffer, 0 for outside) representing the criterion.



(a) Railway



(b) Channel



(c) Road



(d) School



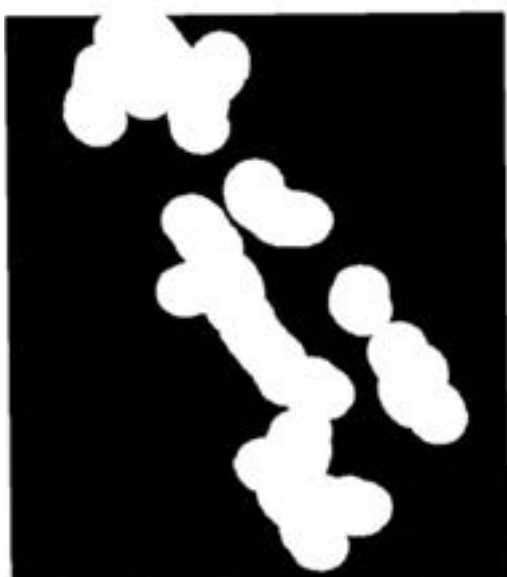
(e) Hospital

RASTER FILL

➤ Explicitly fill the 'No Data' areas with a specific value (e.g., 0).



(a) Railway



(b) Channel



(c) Road



(d) School



(e) Hospital

RASTER REPROJECT

- Raster layers (for LULC) were ensured to have the same Cell Size, Extent, and Coordinate Reference System (CRS) to align perfectly for the final calculation.



Raster Fill for LULC



Raster Reprojected for LULC

Weighted Multi-Criteria Analysis - WMCA

- It combines all the prepared, reclassified raster criteria layers based on their relative importance (weight).
- Each layer is multiplied by its weight, and the resulting layers are summed up to produce a final suitability map, where higher values indicate a more suitable location.

Weighted Multi-Criteria Analysis - WMCA

Weighted Multi-Criteria Analysis - WMCA

Select **Raster**

LULC_reproject

Load

Remove

Enter the **Weight** of the respective raster:

	Raster	Weight
1	LULC_reproject	0.5
2	school_clip	0.1
3	road_clip	0.1
4	rail_fill	0.1
5	hospital_fill	0.1
6	channel_fill	0.1

Instructions:

- Select only single band **rasters**, uploaded to the QGIS Layers, in **.tif** format, with the same noData value in all rasters;

- Rasters must have the same xy and pixel size;

- To remove a created line just select it (or a cell) and press the "Remove" button;

- Enter the weight of each raster in the "Weight" column, with "." (dot) as a decimal separator (eg 0.3);

- The weight must be different from 0 (zero) and the sum of the weights must be equal to 1;

- To group classes from **1 raster**, just assign weight 1 to the raster (on this screen) and assign the same grade (next screen) to the classes to be grouped.

NOTE: Avoid using a raster with a large number of classes (many different pixel values) as loading such values to be reclassified on the next screen becomes time consuming and impractical.

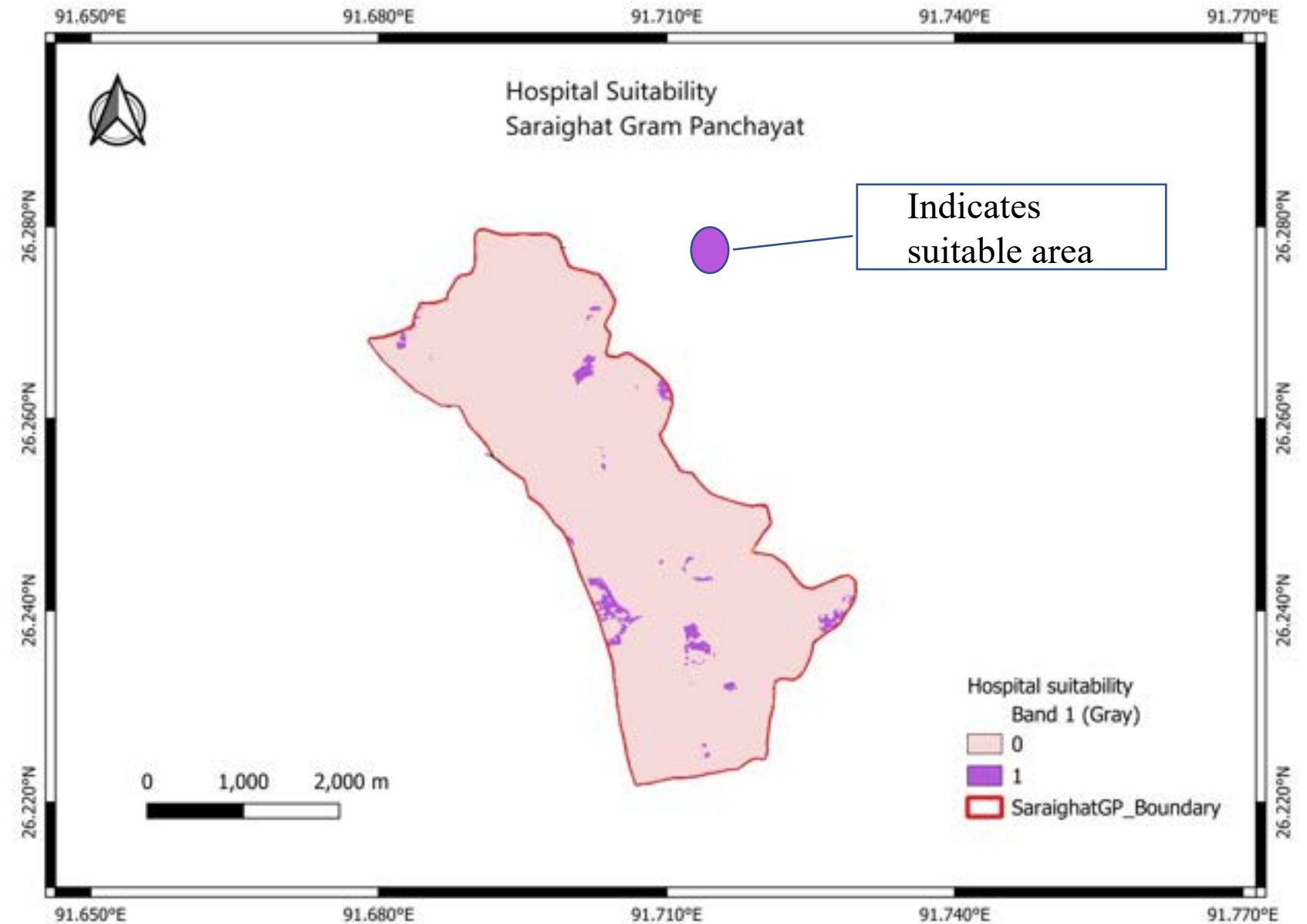
Cancel

Previous

Next

SUITABLE AREA

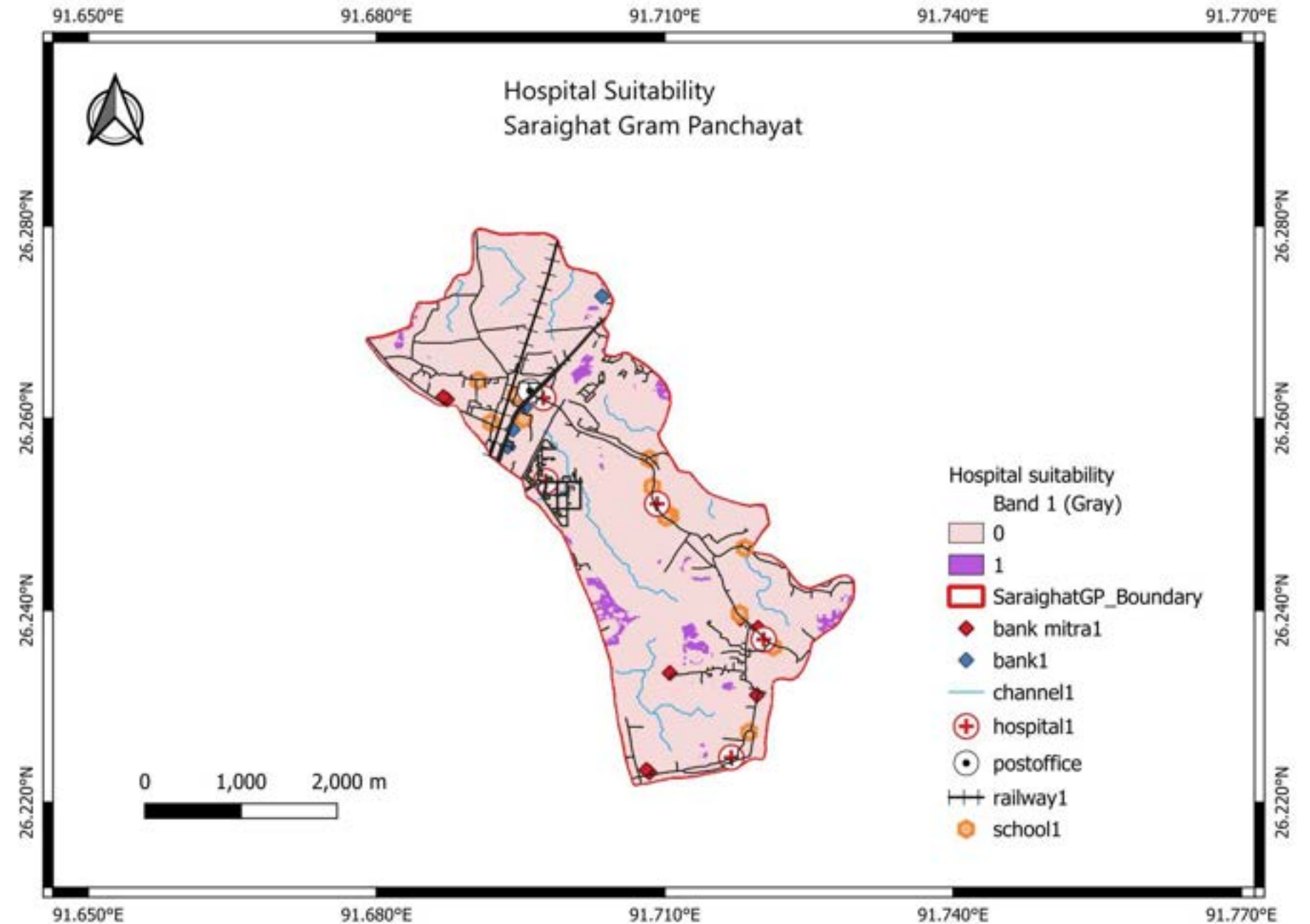
- Clip by mask layer
- The final raster was clipped using the Saraighat Gram Panchayat boundary as a mask layer to show results only within the study area.



RESULT

FINAL LAYOUT MAP

➤ The suitable areas were overlaid with the utility layers for visual analysis and site identification.



CONCLUSION

- The analysis successfully identified multiple suitable land parcels (indicated by purple on the map) that meet all defined criteria and constraints.
- These sites are far from existing hospitals, close to the road network, and avoid major constraint areas like waterways and railways.
- The best proposed site is located.

THANK YOU